

Real Time Ray Tracing in AR

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ABSTRACT

Ray Tracing has played a pivotal role in enhancing the visual realism of computer-generated images, movies, and, more recently, games. It involves simulating the behavior of light rays and has traditionally demanded substantial computational resources and specialized expertise. Despite significant technological advancements in Augmented Reality (AR), the absence of this crucial component has resulted in graphics that lack authenticity.

However, in February 2023, Snap introduced a new version of its AR authoring tool, Lens Studio, which enables developers to seamlessly integrate real-time Ray Tracing into AR experiences, bridging this gap. Snap has become the pioneer in delivering real-time Ray Tracing capabilities to mobile devices across iOS and Android, achieving scalability without relying on dedicated hardware. This accomplishment is attributed to a novel ray tracing pipeline that has emerged from five years of extensive research and development, meticulously tailored to address the resource constraints and limitations of standard mobile GPUs.

The combination of Ray Tracing and other advanced AR tools empowers developers to bring remarkably realistic experiences that blur the boundaries between the digital and physical realms. Notably, industry leaders such as Tiffany & Co. and Disney are some of the brands already embracing the potential of Ray Tracing on Snapchat.

KEYWORDS

Augmented Reality, Lighting, Real-Time, Rendering

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1 CURRENT STATE OF GRAPHICS IN AR

The mobile industry is currently undergoing remarkable growth and advancement in augmented reality (AR) technology, significantly transforming our interaction with mobile devices. With

the increase in power and capabilities of mobile devices, AR applications have gained momentum across many experiences, from gaming and entertainment to retail and even education. Modern AR experiences have witnessed an evolution with the advent of cutting-edge tools and technologies, enabling seamless tracking of surfaces, faces, and complete body movements. This remarkable progress has paved the way for accurate digital object placement in the physical environment.

Mobile graphics have also made significant strides to bring visual quality closer to that seen on computers and consoles. However mobile graphics still fall short, as prevailing approaches rely on simplified physically based rendering (PBR) workflows and basic lighting setups. These approaches result in noticeable disparities when compared to the real world, such as the absence of diffuse and specular reflections, causing digital objects to appear disconnected from surfaces or bodies. The introduction of Ray Tracing technology developed with mobile performance in mind has enabled developers to bridge the remaining gaps and deliver a convincing and visually stunning experience.

2 SNAPCHAT'S RAY TRACING PIPELINE

Snapchat's ray tracing engine is a software-based solution that utilizes the existing GPU functionality found in standard smartphones. To ensure real-time performance, the engine is designed to be flexible and cache-friendly while maximizing GPU occupancy during the tracing process. This approach allows Snapchat to deliver efficient and high-quality ray tracing capabilities on mobile devices without requiring additional hardware support.

Snapchat's new rendering pipeline integrates ray tracing in a series of stages. The process begins with Ray Generation, where rays are selectively generated only from objects marked as "receivers." These rays are then traced only towards objects designated as "emitters". Once we've collected the hits, we move to the Hit Shading stage, iterating over emitters to accurately shade pixels where the generated rays intersect with an emitter. To refine the results, a proprietary denoising filter is applied in the Gathering stage with respect to the attributes of the generated rays. Finally, the Shading stage involves sampling the Hit Shading results for each object marked as a "receiver" and determining their specular or diffuse properties based on the type of ray generated. Additionally, both Ray Generation stage and the Gathering stage can be customized by the user to control the desired effect.

This pipeline not only ensures the seamless integration of ray tracing but also provides users with enhanced control, allowing selective application of ray tracing for improved visual fidelity and realism while maintaining a relatively low performance cost. For

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Figure 1: With the addition of Ray Tracing, digital objects can be anchored to the physical space and appear visually consistent with other objects in the world. Image captured in real-time on an iPhone 14.

previous work on graphics pipeline for ray traced powered AR see [Lira dos Santos et al., 2012].

3 RAY TRACING TOOLS

By harnessing the power of Ray Tracing and optimizing it for mobile devices, we can anchor 3D assets in the real world, resulting in significantly more convincing and immersive graphics. This integration of advanced techniques, coupled with modern advancements in AR technology, introduces a new layer of interactive possibilities while preserving the integrity of the visual language.

While RT alone falls short in achieving content that matches the level of real-world detail, techniques like photogrammetry and intelligent retopology tools enable high-quality assets which completes the desired level of realism. However, these assets often impose a computational burden on accurate ray tracing. To address this, our customizable RT approach leverages optimized lower-polygon versions of the assets, which solely function as emitters and remain invisible in the primary result. This allows us to overcome the computational challenges while preserving the overall visual quality and fidelity of the rendered scenes.

To faithfully simulate the real-world environment and obtain accurate ray tracing data, our approach incorporates a combination of specialized tools. One such tool is our unique reprojection technique, which collects screen data and projects it onto a mesh. This enables the intersection of rays with digital objects, effectively replicating the interaction with the physical world. By employing this innovative method, we can achieve a more realistic and immersive experience where virtual objects seamlessly integrate with the surrounding real-world environment.



Figure 2: Digital objects can now match their real-world counterparts and appear anchored to the world. On the left, the purse without Ray Tracing. On the right, the purse with Ray Traced Reflections. Image captured in real-time on an iPhone 14.

4 ADDITIONAL CHALLENGES FOR REALISM IN AR

Developing AR experiences for mobile devices necessitates working within specific limitations, particularly regarding size and performance. For instance, Snapchat imposes an 8MB total limit on lenses to ensure swift loading and seamless transitions between experiences. This presents a significant challenge when adapting high-end productions for mobile platforms while maintaining the brand's visual quality. High-end production assets often consist of complex meshes and textures, resulting in gigabytes of data and numerous texture maps.

To faithfully recreate the brand experience, intensive optimization is essential. This involves meticulous retopology, reconstructing UVs as needed, consolidating maps into fewer UV sets, and rendering high-resolution shaders into textures. Additionally, by creating separate low poly meshes that serve as invisible emitters that preserve the object's silhouette and key focal points, we can optimize the scene while retaining visual impact.

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- Artur Lira dos Santos, Diego Lemos, Jorge Lindoso, and Veronica Teichrieb. 2012. Real Time Ray Tracing for Augmented Reality. 14th Symposium on Virtual and Augmented Reality, Rio de Janeiro, Brazil, 2012, pp. 131-140, <https://doi.org/10.1109/SVR.2012.8>